Reuse a scene

Digital Technologies · School computers

You will

- make a scene you can use again and again, like a crate, coin or enemy
- put copies of it in another scene
- change the original so every copy updates

You need

- your Godot project open, with a main scene (see the guide Make a Godot project)
- an image for your scene, in assets (see the guide Add images and sounds)

Before you start

- A scene saved on its own is like a stamp. Every copy you place comes from the same original.
- Name scenes in lowercase, like box.tscn, and save them in the scenes folder.

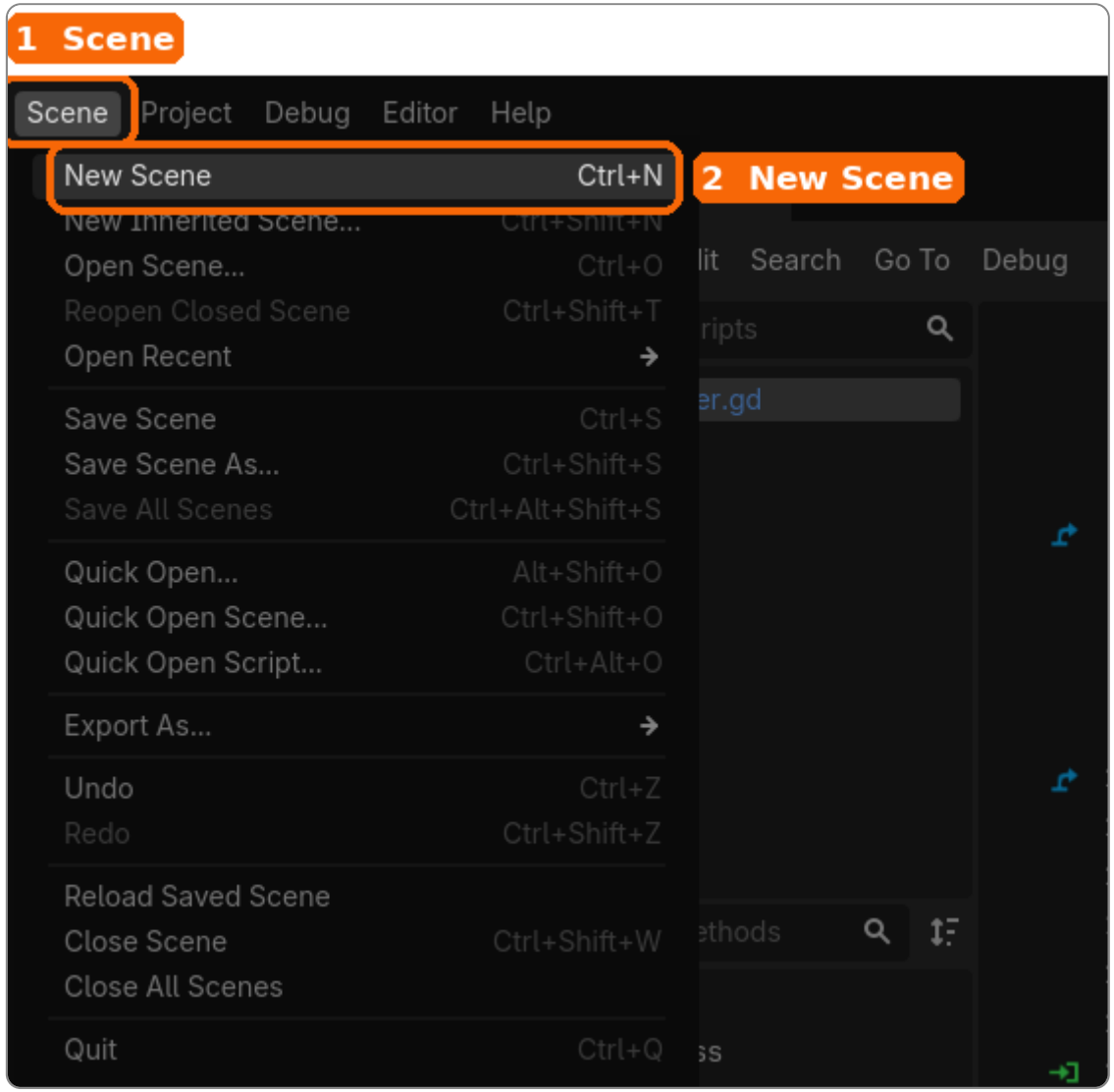
 Click here  Look here

1

Godot › Scene menu

Click **Scene** › **New Scene**.

Or press `Ctrl` + `N`. A new, empty tab opens.

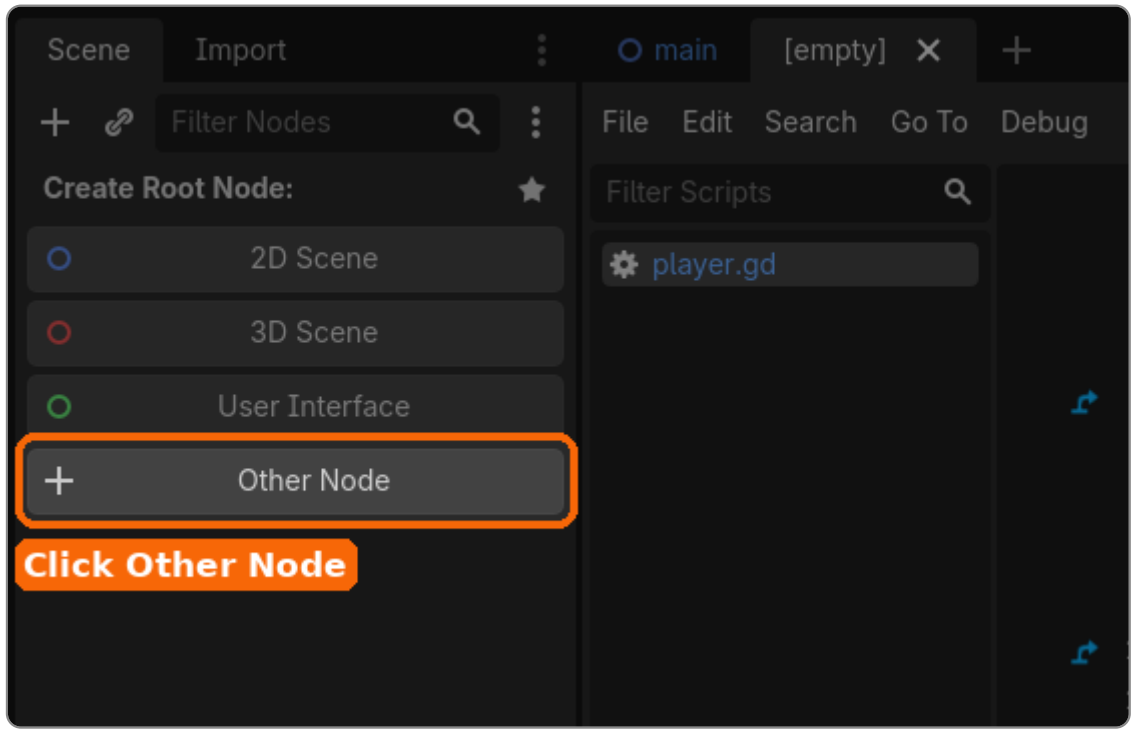


2

Godot › Scene dock › new scene

Under **Create Root Node**, click **Other Node**.

The root is the top node of your new scene.



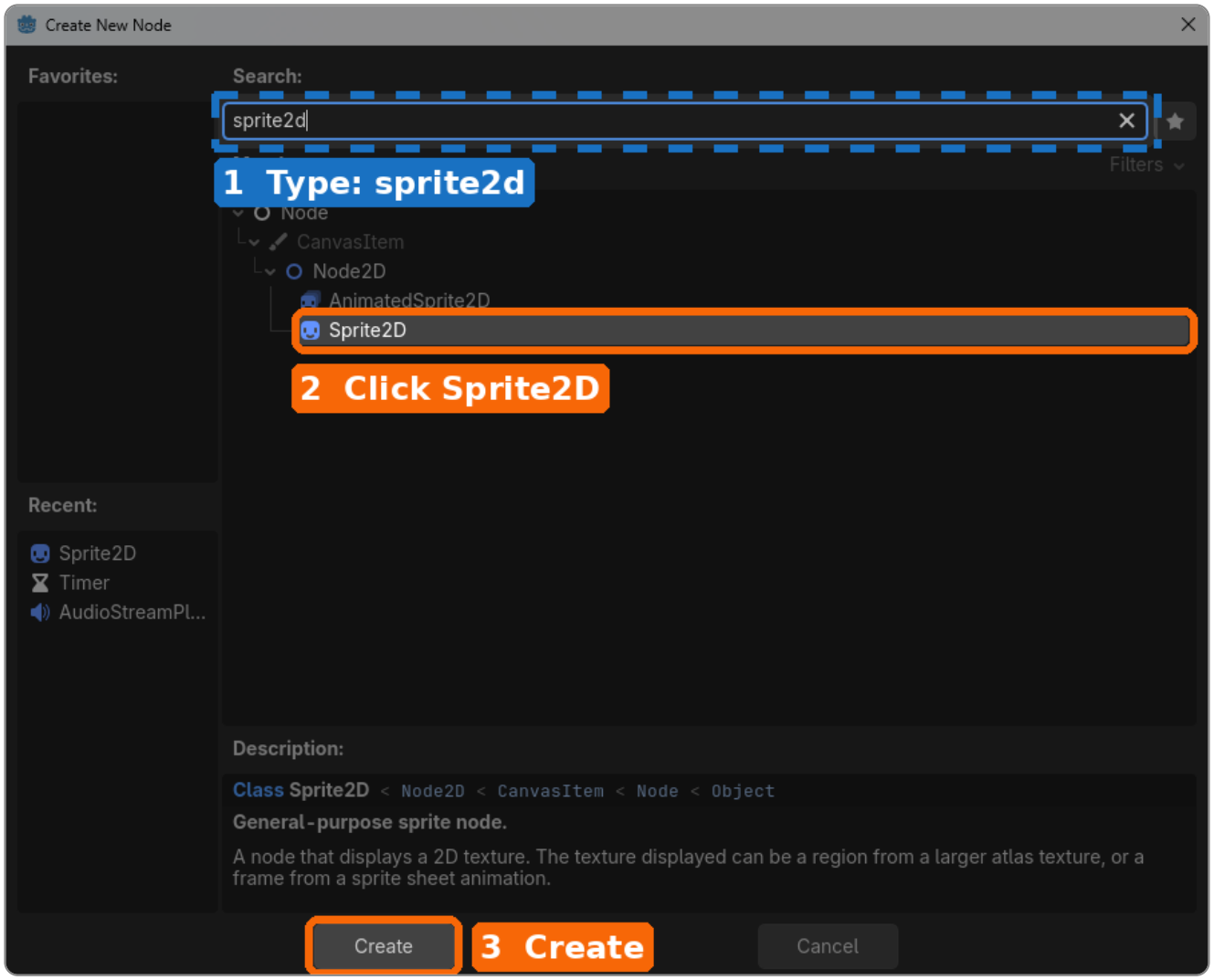
3

Godot › Create New Node

Type `sprite2d`.

Click **Sprite2D**, then **Create**.

A Sprite2D shows one picture. Your kaiako may suggest a different root for things that move.



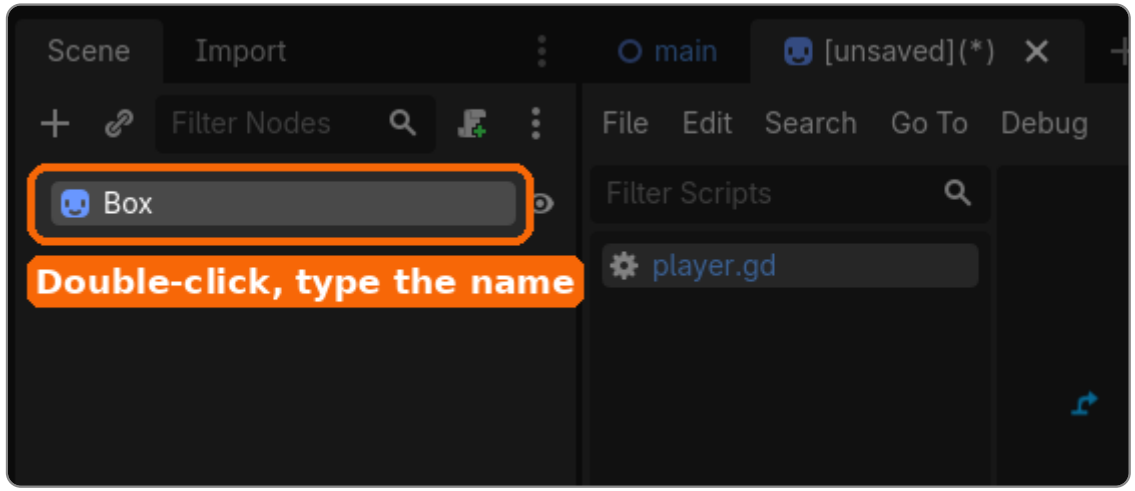
4

Godot › Scene dock

Double-click the new node.

Type a name that starts with a capital (like `Box`), and press `Enter`.

Or click it and press `F2`.

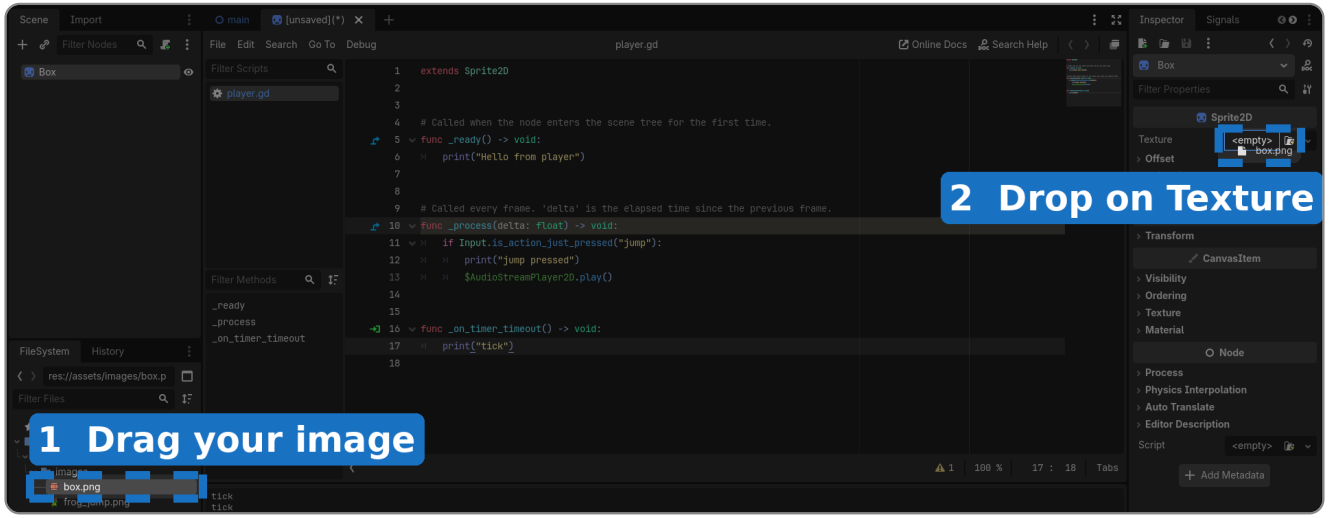


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Godot › FileSystem and Inspector

Drag your image from FileSystem onto **Texture** in the Inspector.

> **Check:** a small picture of your image next to **Texture**.



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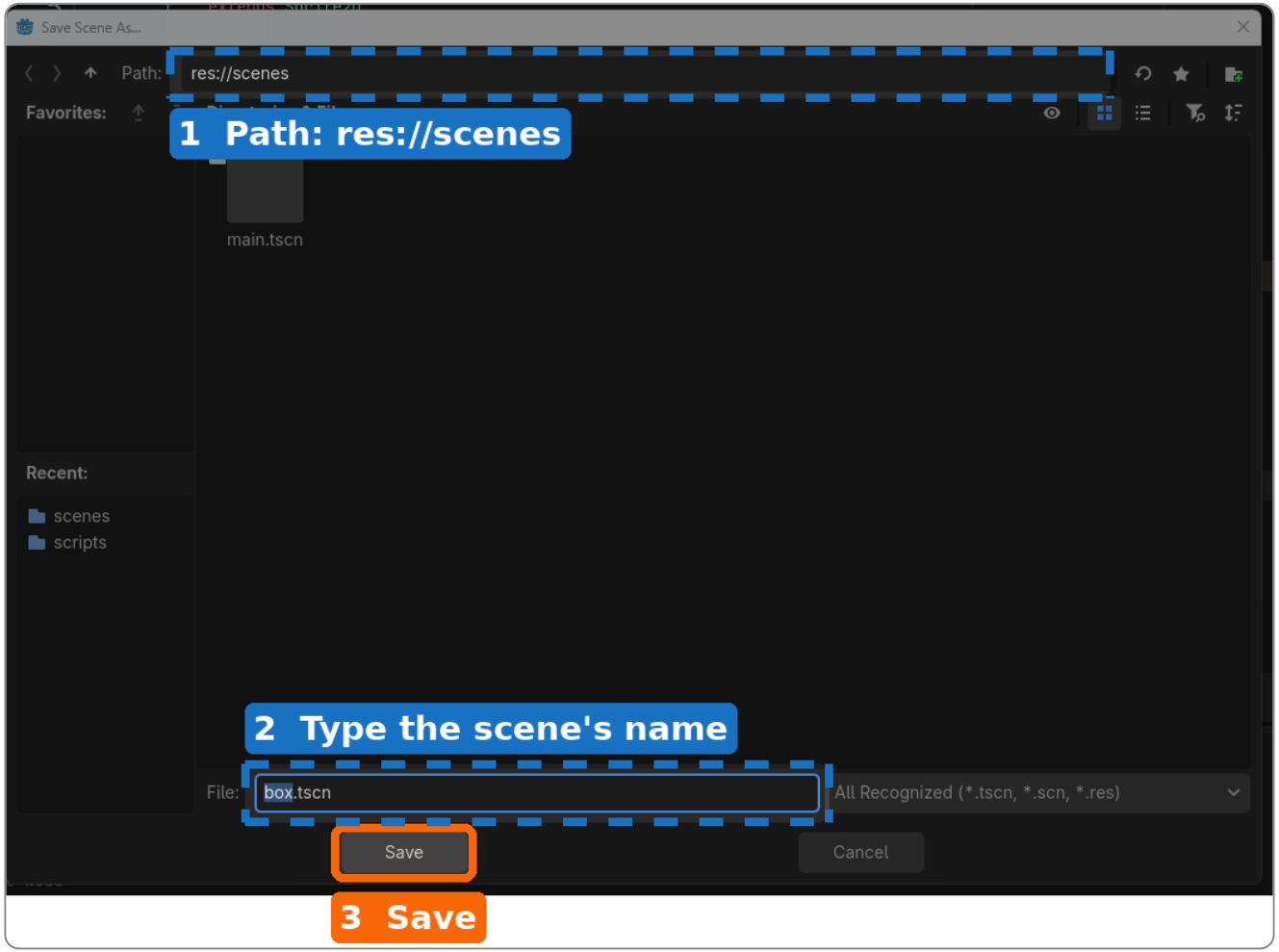
Godot › Save Scene As

Press `Ctrl` + `S` (or **Scene** › **Save Scene**).

Check **Path** says `res://scenes`. Double-click the `scenes` folder if it doesn't.

Check the name, like `box.tscn`, then click **Save**.

> **Check:** a new tab named after your scene, and the file in `scenes` in FileSystem.



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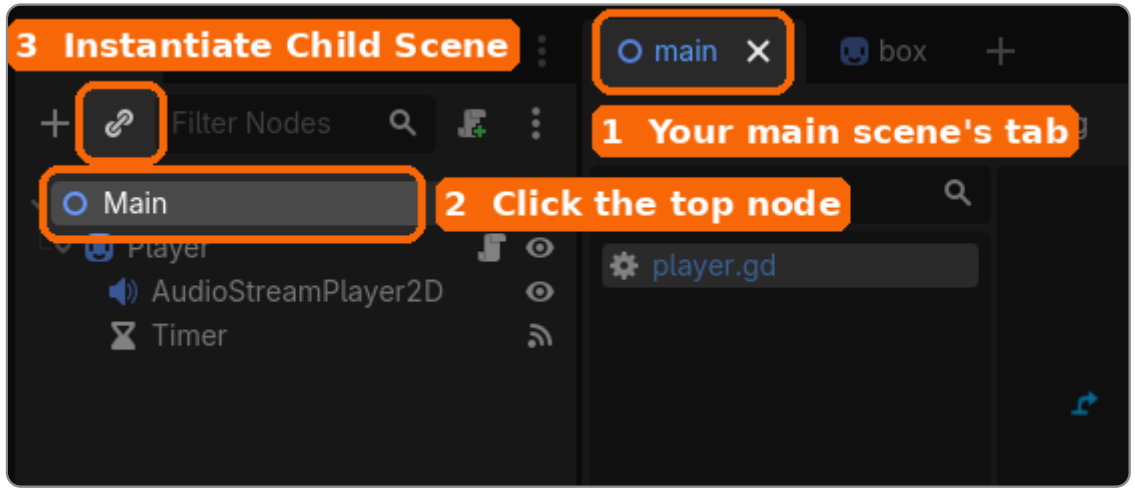
Godot › main scene › Scene dock

Click the tab of the scene you want copies in (here, main).

Click its top node.

Click the chain-link button, **Instantiate Child Scene**.

Or press `Ctrl` + `Shift` + `A`.

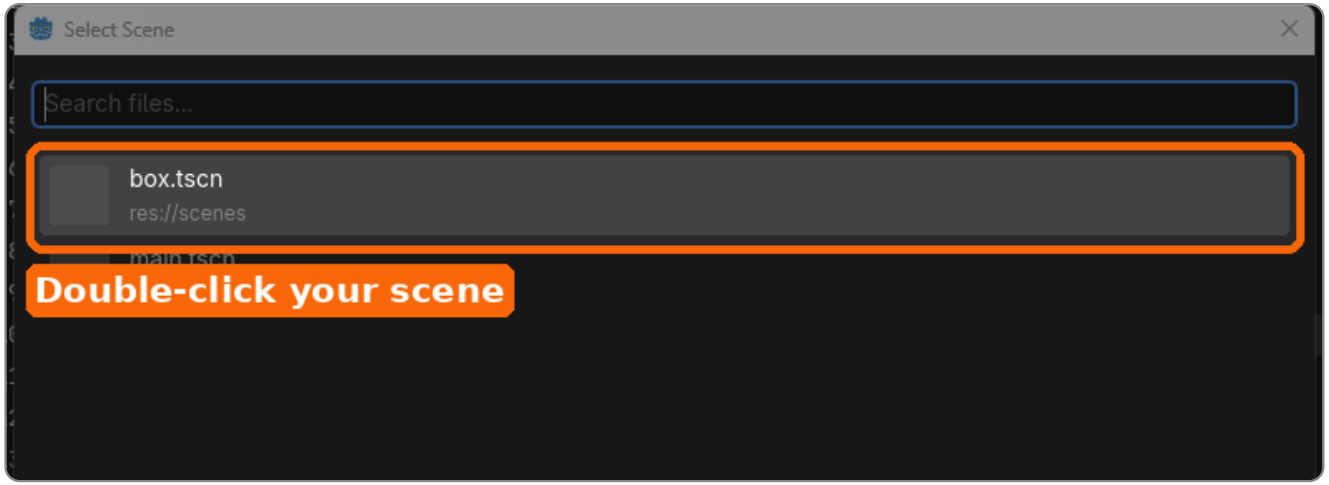


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Godot › Select Scene

Double-click your scene in the list.

Type part of its name at the top if the list is long.



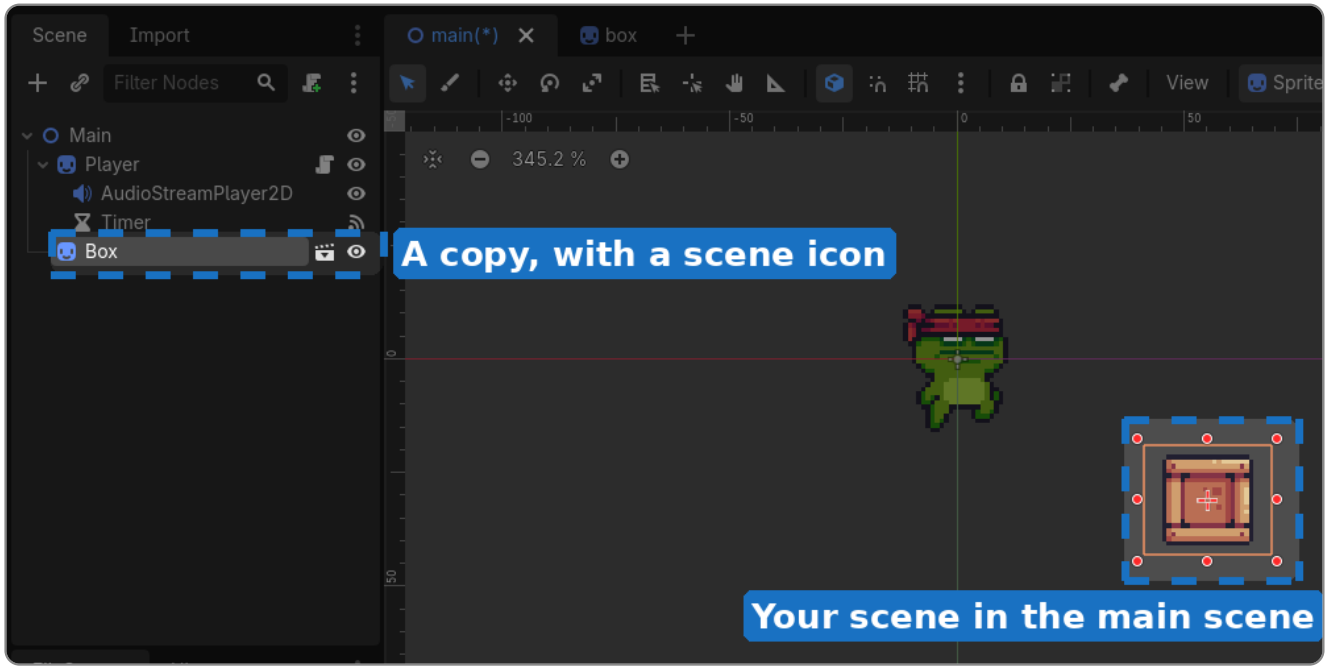
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Godot › 2D view

Click **2D** at the top.

Drag the copy to where you want it.

The scene icon next to it shows it's a copy of another scene.



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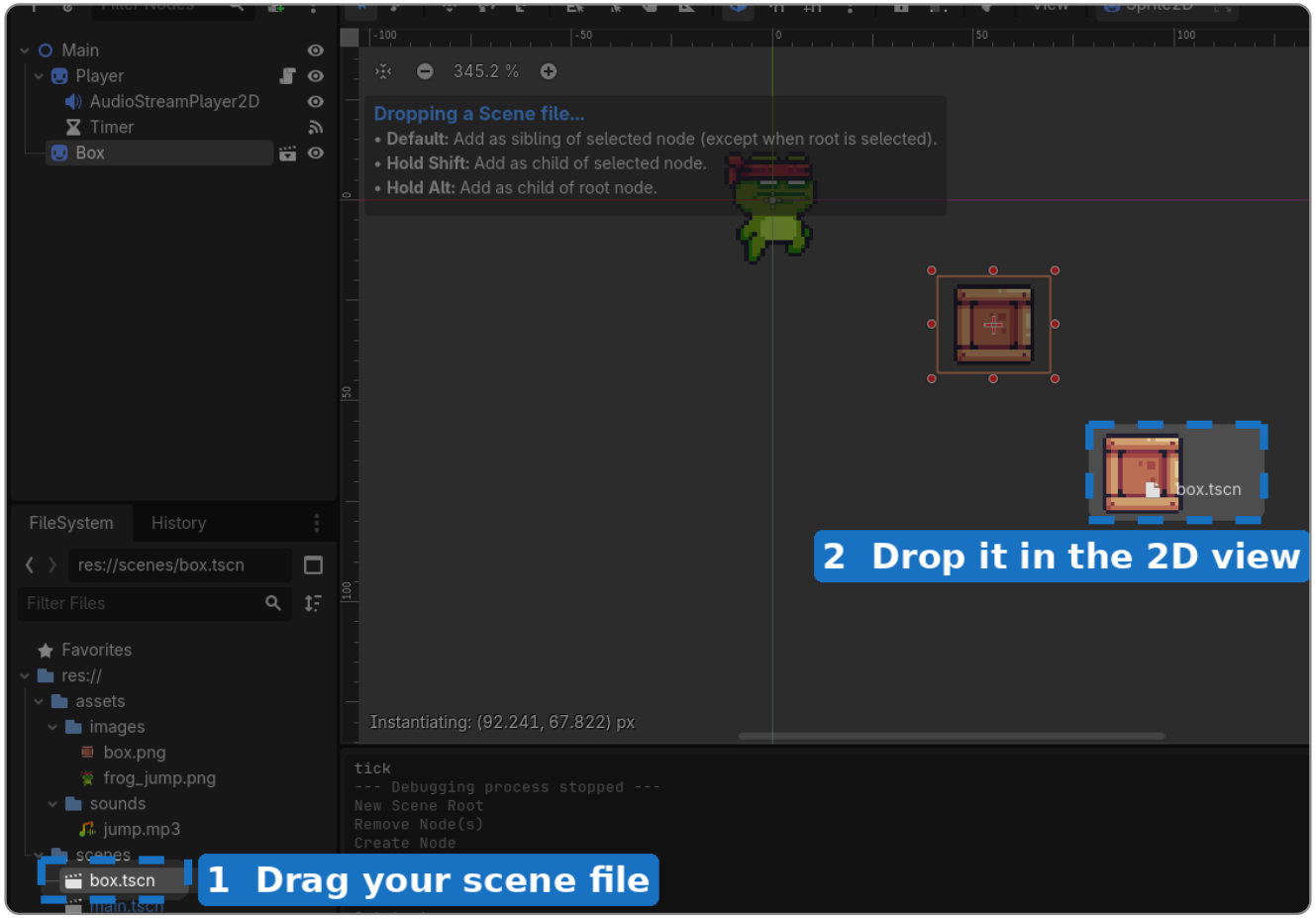
Godot › FileSystem and 2D view

For more copies, drag your scene file from FileSystem into the 2D view.

Do it once for each copy.

Godot names the copies Box2, Box3 and so on.

> **Check:** each copy is listed in the Scene dock with a scene icon.

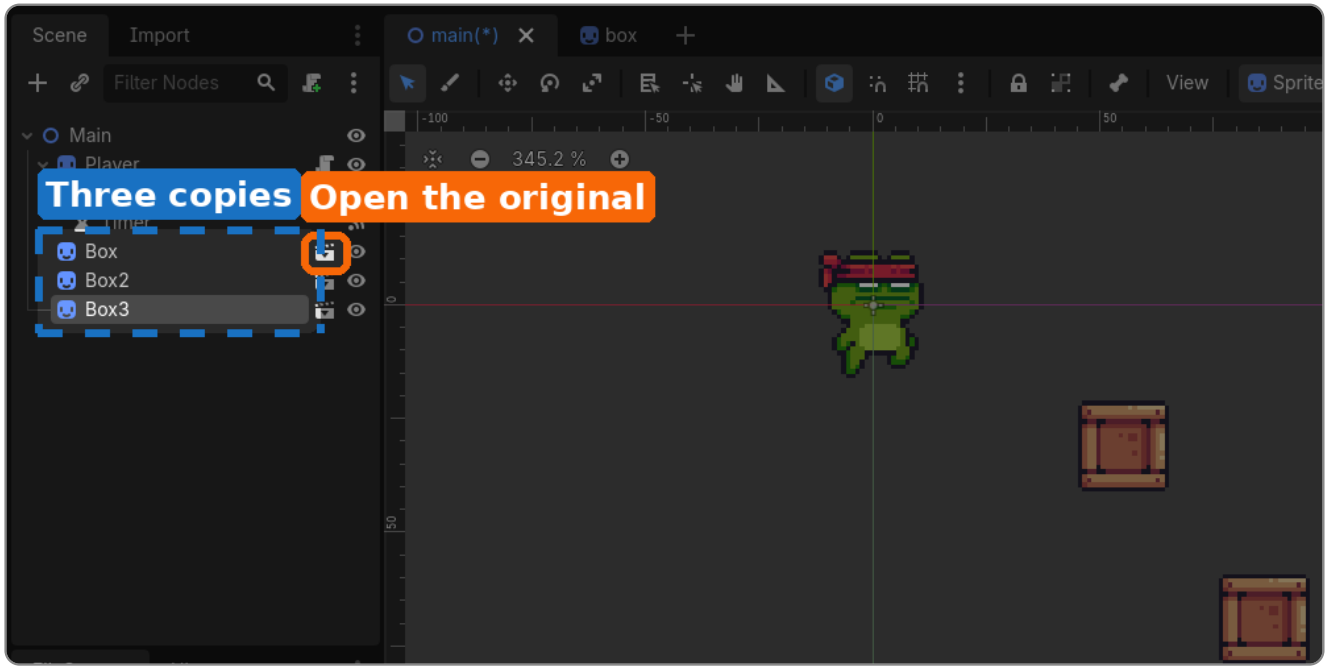


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Godot › Scene dock

To change every copy at once, click the scene icon next to any copy.

This opens the original scene in its own tab. A change to one copy in the main scene changes only that copy.

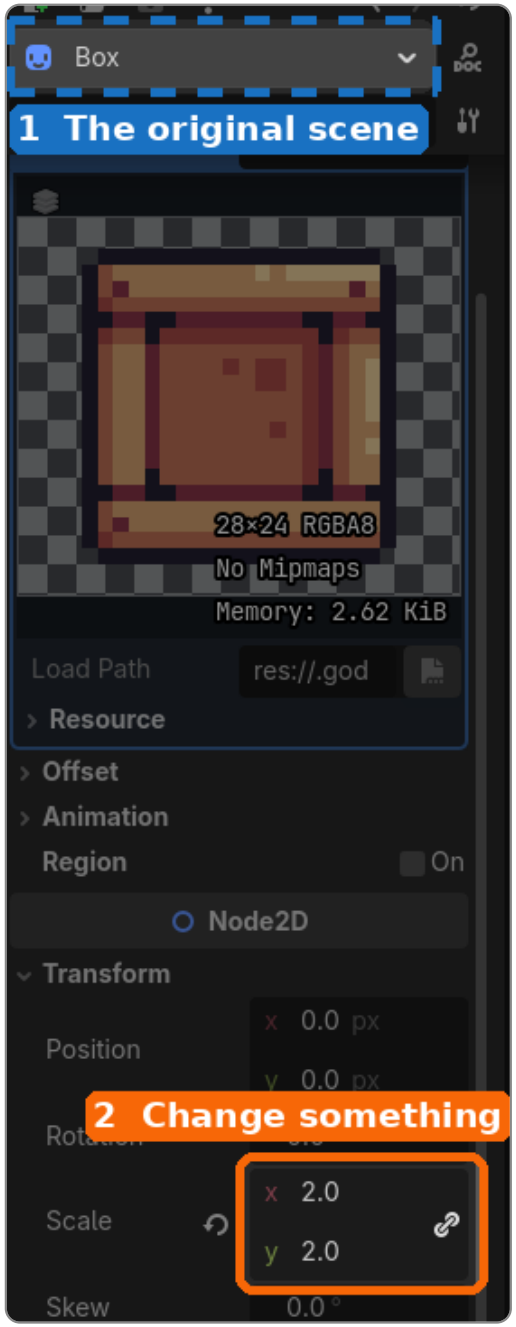


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Godot › original scene › Inspector

Change something, for example **Transform** › **Scale** to `2`.

Press `Ctrl` + `S`.



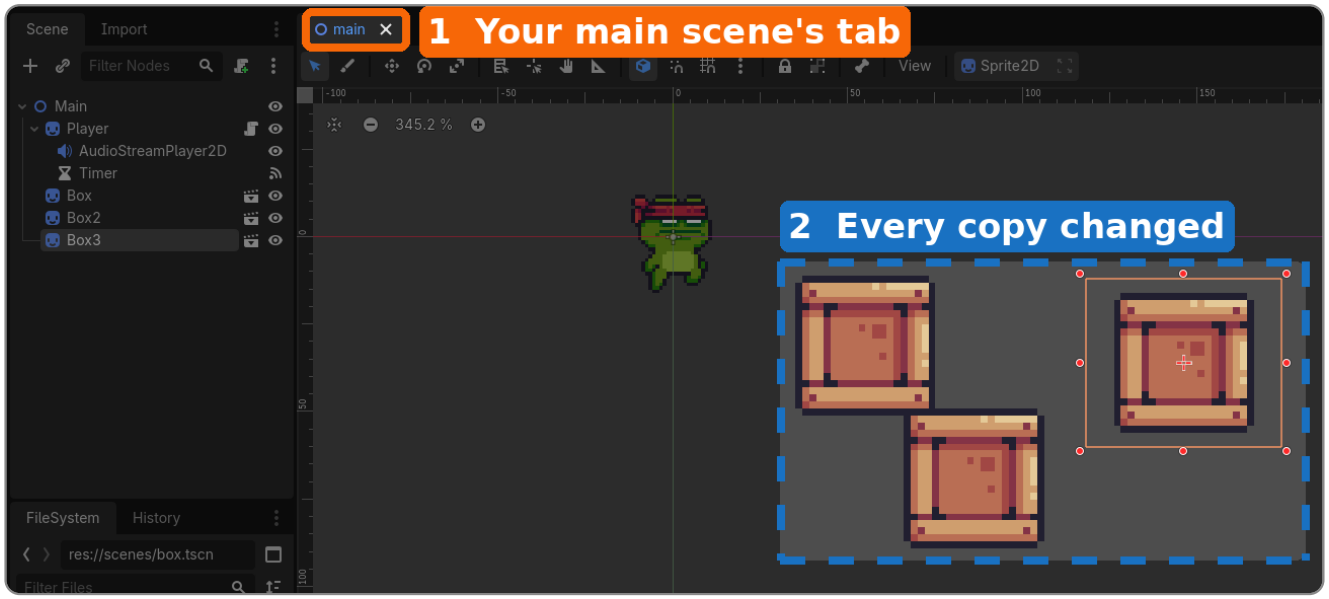
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Godot › main scene

Click your main scene's tab.

Check: every copy shows the change.

If something goes wrong
One copy didn't change: that copy had its own change to the same setting. Click that copy, then click the undo arrow next to the setting in the Inspector.



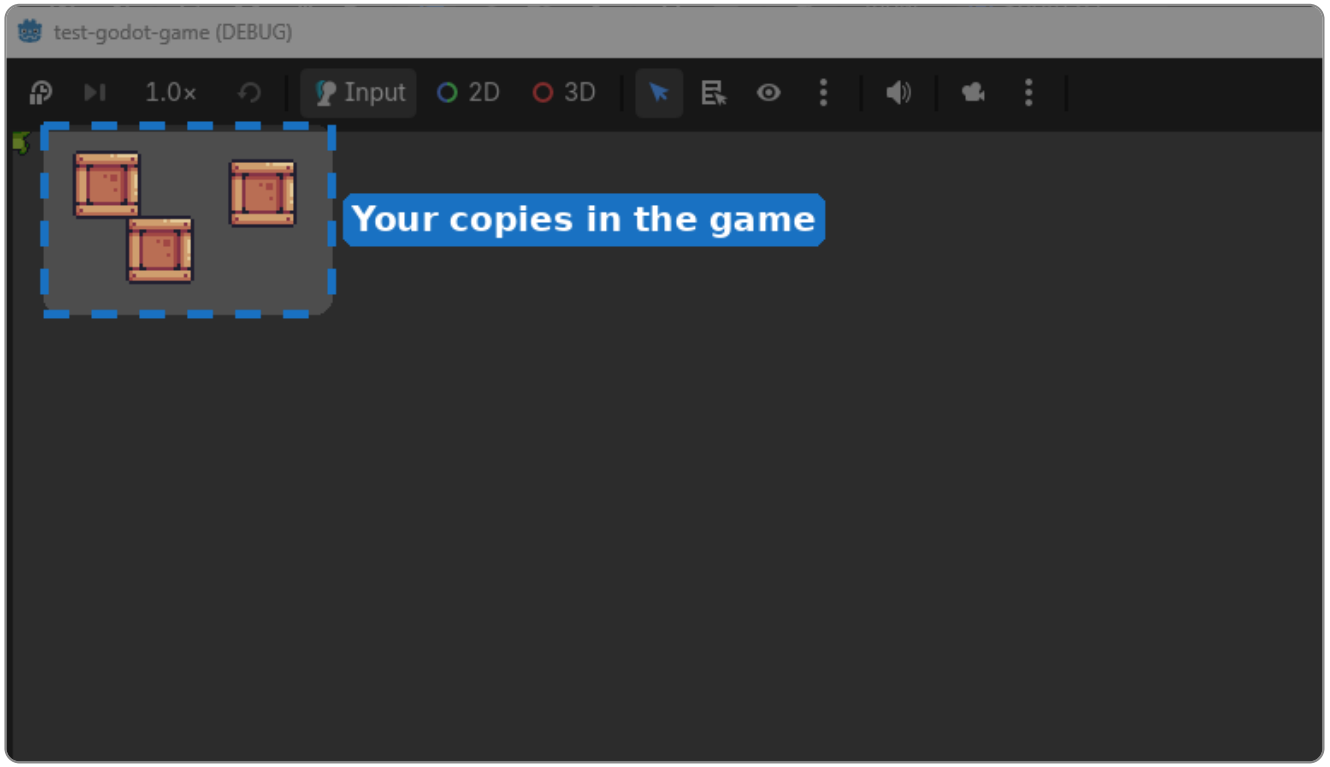
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Godot › Your game window

Press F5.

Press F8 to stop.

> **Check:** your copies show in the game.

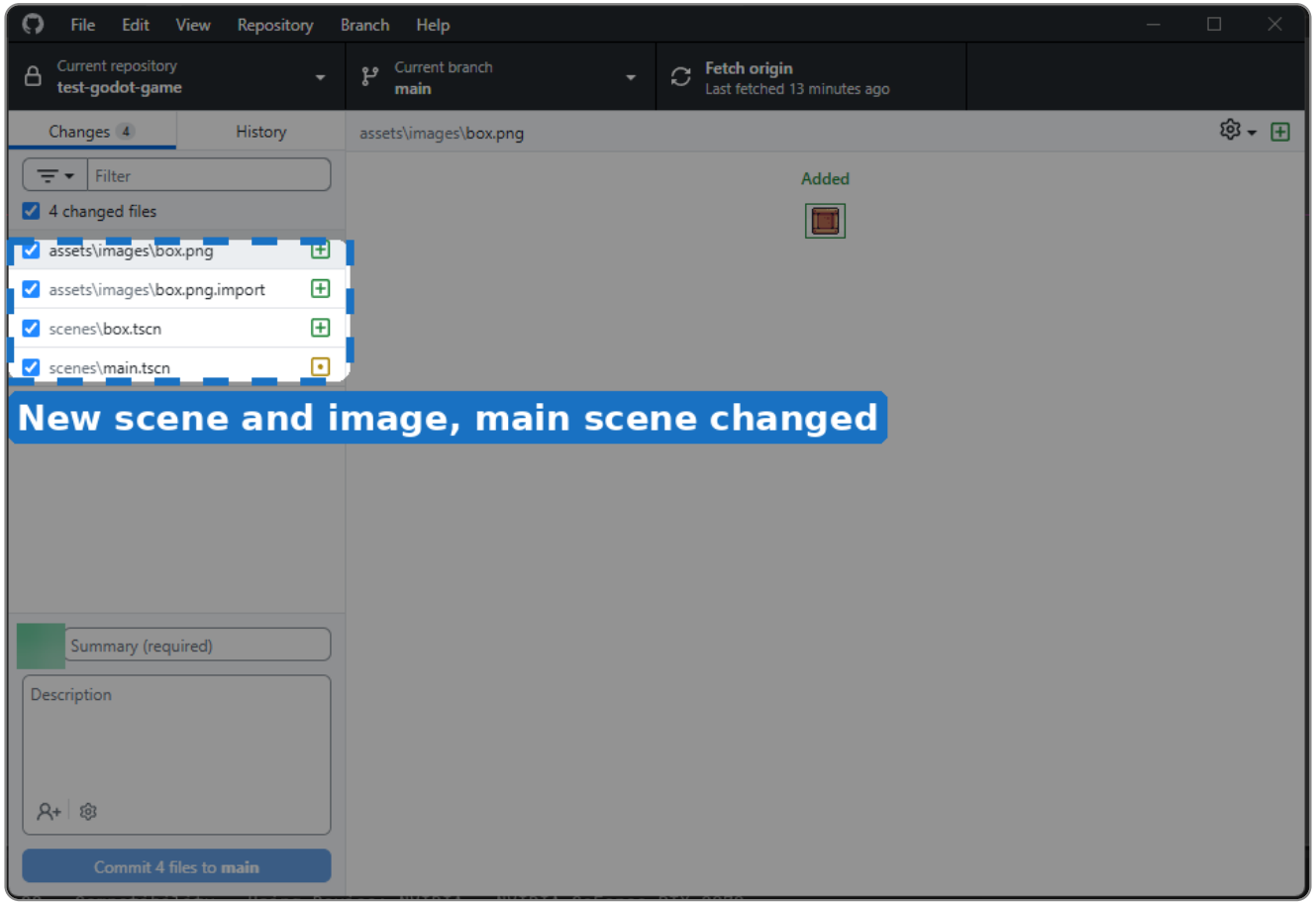


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GitHub Desktop › Changes › if your project is on GitHub

Commit and push with GitHub Desktop, as usual.

Commit the new scene file and the main scene together.



Change the original
To change every copy, open the original scene (the scene icon) and change it there. Change a copy in the main scene only when that one copy should be different.

Good things to reuse
Coins, crates, enemies, platforms, bullets: anything that appears more than once.

Used in this guide
Art: Pixel Adventure 1 by Pixel Frog (CC0), pixelfrog-assets.itch.io.

Checkpoint

- [ ] I can save a node as its own scene.
- [ ] I can place copies of a scene in another scene.
- [ ] I can change the original so every copy updates.